Cognitive Homelessness and Maximizing the Unexpected

#planningformistakes #wandering

Philosophers are often concerned with normative questions about how people should decide to act rather than descriptive ones about how they in fact do decide to act. However, normative decision theory cannot always guide us through important decisions (LA Paul 2015a; 2015b; Carter 2026). An opaque decision problem represents the situation of an agent who must decide what to do while being (potentially) uncertain about their own credences. When confronted with personally transformative decisions, we often have imperfect access to our own epistemic states (cf. Jackson 1991; Howard-Snyder 1997; Lenman 2000). This results in us having uneven evidence in that our confidence towards genuine personal states of value are often higher than our confidence towards illusory personal states of value (Lasonen-Aarnio 2015; Salow 2018; Isaacs and Russell 2023). Therefore, in such cases, it is often easier to be confident that we are in love, when we are in fact in love, than it is to be confident that we are out of love, when we are in fact, not in love. While however, such decisions may not be practically decided on, being in love versus not being in love, they may be narratively decided on (Rust 2025). Using Stump 2010, I will argue that a rational decision theory then may be improved by a narrative decision theory of rational choice.

This results in us having uneven evidence in that our confidence towards genuine personal states of value are often higher than our confidence towards illusory personal states of value (Lasonen-Aarnio 2015; Salow 2018; Isaacs and Russell 2023).

At the center of this debate is the fact that we do not always have the access to our own states and that many decisions require such access. Sam Carter in his recent paper Planning for Mistakes argues that expected utility theories fail for a class of decision problems called opaque decision problems. However, we might think that a narrative use, e.g., of video games, more directly illustrates the problem of uneven evidence in rational choice theory. I argue that Eleonore Stump develops just such an account. Stump defends a normative theory of rational choice based on narrative-self accounts of personal identity.

She argues that analytic philosophy becomes too narrow and myopic, and excludes “complex, nuanced thought, behavior, relations of persons” (Stump 2010, 23). In this account, she

distances her view from more standard ones in rational choice theory. I contrast her account of narrative thought with a more recent approach in rational choice theory. I argue that a narrative self-account of rational choice might include narrative worlds by using possible worlds logic in place of possible outcomes of expected utility theory. In doing so, I attempt a rational decision formula that is both philosophically rich and illuminating.

Normative decision theory cannot always guide us through important decisions (LA Paul 2015a; 2015b; Carter 2026).

How do we decide what to do? Decision theory proposes an answer. When confronted with personally transformative decisions, we often have imperfect access to our own epistemic states (cf. Jackson 1991; Howard-Snyder 1997; Lenman 2000). Standard normative decision theories cannot always guide us through important decisions (LA Paul 2015a; 2015b; Carter 2026).

We might follow some plan of action, only to find out that the promised outcome is not what we desired after all. Such mistakes in judgment then, stem from failures to recognize our own priorities. We call these opaque decision problems. In other words, they stem from errors in judgment about ourselves.

According to some, whether or not these mistakes are avoidable is an open question. This is because of the plausibility that a key kind of evidence necessary to make correct judgments about ourselves, is not a part of the set of evidence that we have. There are two versions of this case however. In the Rothko case, we would have more reason to question whether one or the other is genuine, assuming that we have each of them before us versus if we only see one, then we would have less reason to question whether there is another one that exists somewhere else.

Therefore we assign equal credence to either possibility, that the one we will bid on is real and the one we do not know about is fake or vice versa. Therefore, we might mistakenly think that upon looking at a counterfeit Rothko, that we desire to bid on the counterfeit Rothko rather than a genuine one.

As such, when confronted with personally transformative decisions, we often have imperfect access to our own epistemic states (cf. Jackson 1991; Howard-Snyder 1997; Lenman 2000).

Decision theory derives from a long tradition in consequentialism "If, then, there is some end of things doable in action that we wish for because of itself, and the others because of it" [Nichomachean Ethics Bk I, chapter 2]. The implication here is that we should always choose to do that action that we believe will produce the most value. However, "doing whatever will provide the most value" has been criticized [See Lewis, 1981; also see Carter; 2025; Gallow 2021; and Isaacs *et al* 2023; also see Smith 2010.]. It is argued that we do not have access to the world in a way that would allow us to follow through on our plans to maximize value (especially see, Gallow, J. Dmitri. 2021. "Updating for Externalists." *Noûs* 55 (3): 487–516. https://doi.org/10.1111/nous.12307.). Lewis makes a distinction between causal and noncausal approaches to decision making. Noncausal rational choice theories do not make mention of an agent's credences as a factor in their decision making. Orthodox theory, attempts to improve

on decision theory. We should do whatever action has the most expected outcome (Williamson 2000;2011, cf. Srinivasan 2015; LasonenAarnio 2019;2024; Lasonen 2025;2026; Hughes 2021). In other words, decision theory advices that whatever we decide, that when we act, we should act in a way that maximizes our utility (See Brian Skyrms, "Causal Decision Theory," *The Journal of Philosophy* 79, no. 11 (1982): 695–711, https://doi.org/10.2307/2026547.); also see Isaacs and Russell 2023) or perform the action that leads to the outcome with the most expected value.

In following this course of action, we will often make mistakes however. Sometimes it may be because we are uncertain about which acts, out of those available to us, will have the best outcome. There are various reasons why such access is unavailable. Most notably, this kind of evidence is absent when a value state would be experienced by another who is not us. For instance, historically, human beings have often struggled with attributing the mental states they experience, to other human beings. This is because we are rarely certain about the world around us. We are sometimes unaware of other persons who inhabit the world with us, but even more importantly, we are unaware of who these persons may be. For instance, we also struggle with attributing a certain set of values to future versions of ourselves. Therefore, as the objection goes, our imperfect epistemic access to the state of the world imposes limits on our ability to follow Maximize Expected Value" (Carter 2026, 2; cf. Jackson 1991; Howard-Snyder 1997; Lenman 2000). Our failure to achieve knowledge of other minds, even our own, is an example of this.

In summary, Orthodox decision theory recommends acting in a way that maximizes actual or expected value. However our credences with regard to what we value, are opaque and therefore we cannot identify those acts which maximize actual or expected value (Carter 2026; Williamson 2000;2011, cf. Srinivasan 2015; Lasonen-Aarnio 2019;2024; Lasonen 2025;2026; Hughes 2021). Some decisions require such access however. In later sections, I look at recent attempts to save maximize value. I then defend a narrative self account of rational choice. In this section however, I will offer Eleonore Stump's analysis of suffering as an inability to recognize the value of relationship for creatures like us.

One specific problem in rational choice theory, is the fact that we are often wrong about our priorities. There are many versions of this concern. For instance, let's imagine that on the basis of a coinflip, we will either be shown a genuine or a counterfeit Rothko. If we are shown the counterfeit but not the genuine version, we would be less certain about questioning its authenticity, since we would have no reason to question the authenticity of the fake Rothko having no belief about the real one.

What this means is that in the Rothko case, we have no reason to assume that the one we are bidding on is a counterfeit since it is the only one. This shows a different version of the problem, where we have no reason to value a different state of affairs since there is only one. The kind of common mistake that we make, which is important for Stump's thesis, is that an opaque decision problem with respect to decision theory concerns special personal relationships. If we are confronted with two alternative versions of ourselves, then we would have more reason to question whether one, rather than the other, is genuine. Stump argues that narratives give

us reason to question whether one life, rather than any other, is genuine. I think that the best example of this fact can be represented with the following lyric excerpts from the songs "By Yourself" by Tyrone Griffin Jr (Ty Dollar $ign Diggs) and "Rich and Sad" by Austin Richard Post (Post Malone).

4 Narrative and the Knowledge of Persons

Can an agent always expect to do best by trying to do whatever has the most expected value?

- Orthodox decision theory recommends acting in a way that maximizes expected value,
  - do whatever has the most expected value:
    - i.e., [be] "your own boss"

According to Orthodox decision theory then, agents like us should follow the condition on their credences which is both necessary and sufficient for maximizing expected value (MEV). However, Orthodox decision theory is not always good advice for agents like us. Agents like us are susceptible to error. For agents like us then, who are susceptible to error, there can be situations in which trying to do what has the most expected value (follow MEV), will leave us worse off. Imagine that the following represents: Doing whatever has the most expected value for a given agent

- "Stacked my bread and bought my own Mercedes (Vroom, vroom). I'm my own boss, do it my way (Way)

and then getting it wrong

- "All this stuntin' couldn't satisfy my soul (–oul), Got a hundred big places, but I'm still alone (–one)"

Therefore, while such decisions may not be practically decided on, being in valuing a friendship versus not valuing a friendship. Given our susceptibility to error, it is then important to show which framework works best for evaluating advice. Following Stump, I argue that they may be narratively decided on (Rust 2025).

The Nature of Love and Suffering

Using Stump 2010, I will argue that a rational decision theory then may be improved by a narrative decision theory of rational choice. Shame is the connection between moral wrongdoing and some defect of the will. Shame undermines our ability to experience fruitful and deep relationships with others. If I am guilty of stealing from a friend out of necessity, it will be difficult for me to face that individual and our relationship will be strained. If I desire union with the other person, then I will feel shame given that I expect rejection and abandonment by them. Over time however, I may not feel shame any longer. This is because I will no longer desire a relationship with that person.

The remedy may be to repair the will, but sometimes it is not possible for a person's will to change. Notice here that the temptation to steal from the friend came by way of perceived necessity, and therefore outside of my control. This highlights the difference between shame and guilt. I will feel guilt out of my anticipation to be the object of anger or punishment whereas I experience shame when I am anticipating abandonment or rejection. I experience shame out of my desire to be loved by others and conviction that something about me warrants their rejection of me.

The remedy for shame rests in the need of paired processes of justification and sanctification.

Importantly, an account of friendship for psychological well-being, where this friendship is grounded in union with God.

How do we determine the meaning of concepts like love? A problem with this account, is that it rests on theological commitment. However, it in itself is not sufficient for justifying theistic beliefs. Consider the problem of evil.

1. There is suffering in the world.
2. There is an omniscient, omnipotent, perfectly good God.
3. There is no morally sufficient reason for an omniscient, omnipotent, perfectly good God to allow suffering in the world.

What is bad about suffering is that it undermines or destroys what the sufferer centrally cares about * her own flourishing * the desires of her heart * or both #### Suffering and Pain

Suffering is not necessarily pain, e.g., when a mother chooses to give birth naturally. But what about heartbreak? Some evils are evil because they break the will of the sufferer.

Suffering could be related to flourishing: Having your purse snatched is bad even if there was nothing in it because it undermines your will (the desires of the heart).

However, not all evils violate a person's will. A torture victim might will a slightly less horrendous form of torture. So part of the way someone is diminished by evil is that it makes them acquiesce in their own suffering. However, an even more egregious form of suffering is one where a person is unable to recognize that they are suffering.

Two notions of caring * What we care about * What we are conscious of caring about

Consider the following vignette:

> a college athlete who is a runner at some undistinguished small college and who wants very badly to be the best runner his school has ever had. If we ask him whether he wants to be a mediocre runner, he will surely not say 'yes'. (p. 9)

Another example the English poet John Milton strongly desired to be politically active "in the Puritan government of his day", but the death of Cromwell was a catalyst in his removal from office. Arguably, being a poet is more indicative of Milton's flourishing than being a minor government official, yet, being a minor government official was the desire of Milton's heart.

Being Happy

Neuroscience has discovered how closely related our emotions are to our thoughts. They are indistinguishable in our autobiographical memory. (Becoming Good News, Jessie Cruickshank and Julia Schmaltz, 2026).

Some other related materials, Wandering in Darkness, Eleonore Stump 2010, for instance, questions about what we care about may be autobiographical, "And, on the other hand, there is a person's flourishing as the best he can be.

Being in a flourishing condition is also what a person cares about, even if he does not know what exactly flourishing is for him" p. 9;

and "I will argue that presence and closeness of the sort needed for the union of love required internal integration of the psyche . . . that a self cannot be integrated around just anything; it has to be integrated around the good." p. 109.

Defended by this claim: "The main aim of this chapter is the exposition of Aquinas's account of sanctification and justification, which he takes to be the remedies for the psychic sickness whose source is the human propensity to evil." p. 155.

My interests have largely been moral epistemological. Therefore I am interested in how close friendship is important for moral reasoning, including using reason to discover moral truths and act in morally admirable ways.

Normative decision theory cannot always guide us through important decisions (LA Paul 2015a; 2015b; Carter 2026). #### Standard Decision Formulas

Determining the expected utility of some action

		states	states
		it rains	it does not rain
acts	take umbrella	encumbered, dry	encumbered, dry
acts	leave umbrella	wet	free, dry

The expected utility of an act A (for instance, taking my umbrella)

depends on two features of the problem: * The value of each outcome, measured by a real number called a utility. * The probability of each outcome conditional on the act. * roughly, how likely it is that outcome will occur, on the supposition that the agent chooses act A

$$EU(A) = \sum_{o \in O} P_A(o) \cdot U(o)$$

When confronted with personally transformative decisions, we often have imperfect access to our own epistemic states (cf. Jackson 1991; Howard-Snyder 1997; Lenman 2000).

Therefore we must often make decisions under conditions of uncertainty.

Here is an example: - "Stacked my bread and bought my own Mercedes (Vroom, vroom). I'm my own boss, do it my way (Way)

Rational choice theory says "choose the act with the highest expected utility." This results in us having uneven evidence in that our confidence towards genuine personal states of value are often higher than our confidence towards illusory personal states of value (Lasonen-Aarnio 2015; Salow 2018; Isaacs and Russell 2023).

While however, such decisions may not be practically decided on, being in love versus not being in love, they may be narratively decided on (Rust 2025). #### Response Account of Love

> Love is a response of the lover to qualities he perceives and values in the beloved.

1. Love = x ϕ's y,
    1. x is ϕ with/of/about y
2. x feels emotion e, towards y, then x believes y to be ϕ. $[(\exists\phi(x)(y)(Exy \rightarrow Bx\phi y)]$
    1. For some quality ϕ
    2. x Believes of y that y possesses quality ϕ.
3. if x feels the emotion in question towards some object y then x believes y to have some determinate quality ψ, which normally but not necessarily he will be able to specify: $[(x)(y)(\exists\psi)(Exy \rightarrow Bx\psi y)]$
4. 1-3 are necessary but not sufficient conditions. Justification of e is also needed.

Is love ever unjustified? 1. if x loves y then x wants to benefit and be with y etc. Because x believes y has some determinate characteristics ψ in virtue of which he thinks it worth while to benefit and be with y. 2. Justification (i) the belief that y is ψ may be well- or ill-founded. (ii) the distinction between want and wish is that want is accompainied by the belief that the object is obtainable. (iii) one can be mistaken in attaching the value he does to what he wants.

Augustinian conception: failure to seek help, possible because of God's grace.

But what is sanctification? However, willing the good is related to self-love, love with self and with God. higher-order desire to will the good but presupposes general higher-order desire Faith-unites belief and faith

1. The will can detest its past wrongdoing and love God's goodness
2. The will can cleave to its past wrongdoing and reject God's goodness
3. The will can simply be turned off in regards to its past wrongdoing and God's goodness (will neither refuses nor accepts grace)

Activity versus passivity

Frankfurt: So a person's true self is characterized by desires and volitions, that she cares about, what she cares about are things that are active for her.

However

- Passivity is in some ways essential to the human experience
  - Our belief that the sun is yellow depends on sense impressions that we have no control over.
- Union and presence then also requires passivity, but also activity.

Step 1

An opaque decision problem, $\hat{\Delta} = \langle \mathrm{S}^{\hat{\Delta}}, O^{\hat{\Delta}}, A^{\hat{\Delta}}, P^{\hat{\Delta}}, v^{\hat{\Delta}} \rangle$, where we may be faced with some momentous life decision.

and

A set of worlds, Ω. Following Lewis (1982), is a set of possibilities wherein some agent weights some outcome as valuable or not. Accordingly, there is some world W at which an agent has a credence $C(W)$ that measures the agent's belief at W, and measures how satisfactory it seems to that agent for W to be the actual world.

In this case, there are various worlds which must be taken into account for what we will call an opaque decision problem. We shouldn't yet assume that the acting agent has access to more than one of these just yet. Some worlds will be actual and others possible, hypothetical, and perhaps counter-factual (what could have happened *but did not*). However, the counter-factual worlds here are of no consequence since such worlds would not factor into the agent's decision making in a non-trivial way. But at the moment of the agent's decision, it seems that some decision plan would contribute to a given outcome, and that that outcome would be satisfactory to that agent. Either way, each world contains states and outcomes.

- A set of states, S. States, $s, s', s'', ... \in S$ are objects in a decision problem relevant to the agent's decision but that they maybe unsure about. For instance, whether the agent is in s *in love* or s' *not in love*. Outcomes, $o, o', o'', ... \in O$ are objects of an agent's desires: E.g., "Being your own boss". Acts, $a, b, c, ... \in A$ are the immediate objects of decision-making, "stacked your bread".
- A set of acts, $\mathrm{A} \subseteq S \to O$. A set of acts would include *stop smoking to save money.*
- A state-dependent probability function, P: $S \to (P(S) \to [0,1])$. Here we have a function that tells us the agent's credence at each state. Lewis does not give wait to an agent's credence about who are where they are in a given state. Instead of propositioning that there are two distinct persons inhabiting the same world, there is only one credal possibility comprising of beliefs and values at each distinct world (Lewis, David. 1979.

"Attitudes De Dicto and De Se." *The Philosophical Review* 88 (4): 513–43. https://doi.org/10.2307/2184843. Lewis, David. n.d. *Causal Decision Theory.*).

- A value function, v: $O \to \mathbb{R}$. Which tells us how well the agent takes things to be going at each outcome.

Step 2

The set of state-dependent probability functions represent an agent's credences at different states from the agent's perspective from within the decision problem after potentially acquiring some information about the state they are in. Therefore, for each $s \in S, P_s$ is the credence distribution adopted by the agent at s.

For any opaque decision problem $\hat{\Delta}$:

$\Pi = S \to A$ is the set of plans determined by $\hat{\Delta}$.

Plans, $\pi, \pi', \pi'', ... \in \Pi$ are functions from states s, to acts, i.e., $\pi(s) = a$ is the act a recommended by plan π in state s. MEV (Maximize Expected Value). Let $\Pi_{|MEV}$be the set of plans for MEV in $\hat{\Delta}$. Let $ev_s(a)$ be the expected value of some action a at state s (for any $s \in S$ and $s \in A$).

$$ev_s(a) = \sum_{s' \in S} v(a(s')) \cdot P_s(s')$$

Then $\pi \in \Pi_{|MEV}$ iff for all $s \in S$ and $a \in A$:

$$ev_s(\pi(s)) \geq ev_s(a)$$

The expected value of an action at some state s, is the sum of values of the outcomes produced by that action at each state. We weight the sum of values according to the probability that is assigned to that state from the perspective of that state. A plan to maximize expected value then recommends, in each state, one of the acts which has optimal expected value at that state.

Step 3

The way an agent acts depends on the plan they are trying to follow. However, agents like us who are susceptible to error do not always act according to what their plan recommends for the state they are in. Rather we can understand their susceptibility to err in terms of their underlying pattern of dispositions. To represent these dispositions, we need to represent the state an agent is acting as if they are in, what we can call "enriched state spaces".

For any opaque decision problem $\hat{\Delta} : [\cdot]$ is a bijective function with domain S which is a product of its domain and codomain, $\bar{S}_{[\cdot]} = S \times \{[s] : s \in S\}$

Where any state $s, [s]$ represents an agent whose act is determined according to what their plan recommends for s and an enriched state, $(s, [s'])$ represents the circumstances of an agent who is in s but whose act is determined according to what their plan recommends for s'. Sets of states in the enriched state space can be thought of as propositions which can tell us about:

1. what the world is like
2. what the agent will act as if the world is like

Step 4

Now we musts introduce a state-independent probability function over the enriched state space. This will represent the agent's credences prior to finding themselves in the decision situation.

Definition 5 (State-Independent Probability). For any $\hat{\Delta}$ and $[\cdot]$: a state-independent probability function, P, is a probability function over $P(\bar{S}_{[\cdot]})$.

The state-independent probability function takes a stance on both what the world is like and how the agent will act as if the world is like. But it does not incorporate any information the agent acquires once they are at a particular state in the decision problem. For each state s, it reflects the agent's prior credence regarding both how likely they are to find themselves at s, their prior dispositions for s, and how likely they are to act as if they are at s.

Therefore, for any s, s',

$$P([s']|s) = \frac{P((s, [s']))}{P(\{(s, [s'']) : s'' \in S)\})}$$

That the probability that an agent's estimate of their susceptibility to error at a state, s, relative to a plan, π is their credence assuming that they are in s, that they will act according to what their plan recommends for s' instead.

In other words, "Stacked my bread and bought my own Mercedes (Vroom, vroom)." "I'm my own boss, do it my way (Way)"

[as such] "Got a hundred big places, but I'm still alone (–one)"

[therefore] "All this stuntin' couldn't satisfy my soul (–oul)"

According to P, an agent's susceptibility to error at s, relative to π is proportional to:

$$\sum_{\substack{s' \in S \\ \pi(s) \neq \pi(s')}} P([s']|s)$$

Stunted Soul and **Stackin' Cars**.

Using Stump 2010, I will argue that a rational decision theory then may be improved by a narrative decision theory of rational choice.

- "Stacked your bread and bought your own Mercedes (Vroom, vroom). You your own boss, do it your way (Way)
- "All this stuntin' couldn't satisfy my soul (–oul), Got a hundred big places, but I'm still alone (–one)"

I will argue that these songs are an example of the claim that the lacking evidence in normative theories of rational choice is evidence to the contrary. I then propose that Professor Stump's solution rooted in narrative is sufficient for meeting this challenge. I then provide a rational choice theory grounded in narrative choices.

Evidence to the contrary is the kind of evidence a person needs that what they desire is not in fact what they desire. A special component of this, is the fact that the evidence we need, evidence to the contrary, is not always available to us.

Consider:

> Case: Cumin & Caraway Suppose that you reliably outperform your level of confidence when it comes to identifying spices. For example, you regularly underestimate your ability to discriminate cumin and caraway seeds by sight. Shown a sample of either, you will be wholly uncertain which you are looking at—you will assign equal credence to each. However, forced to choose, you do systematically better than chance at selecting caraway when trying to select caraway, and selecting cumin when trying to select cumin—in each case, you select the spice you are trying to select around 75% of the time. Moreover, you are certain that you are like this. [Carter, Sam. 2026. "Planning for Mistakes." *Noûs* n/a (n/a): e70043. https://doi.org/10.1111/nous.70043.]

We can imagine some agent that wants to believe some proposition p about themselves, for instance that they are in love, but for all p, -p is not a part of the agent's evidence. For all the agent knows, desires consistent with not being in love either are, or are not, a part of their evidence set, not accessible to them as evidence. In this case, an agent has evidence consistent with being in love, solely for the reason that evidence consistent with not being in love, is not a part of their evidence set. This is because they may not have access to the necessary counter evidence, that they are, in fact, not in love.

Offered a bet at 2:3 Odds

$$\begin{array}{rl} \text{Decline your choice:} & 0 \\ \text{Accept, Guess Cumin:} & \begin{cases} \$0.67 & \text{if cumin;} \\ -\$1 & \text{if caraway} \end{cases} \\ \text{Accept, Guess Caraway:} & \begin{cases} -\$1 & \text{if cumin;} \\ \$0.67 & \text{if caraway} \end{cases} \end{array}$$

Maximize Act Expected Value: Decline bet in both states since expected value is -$.17

Maximize Plan Expected Value: plan to accept cumin if cumin and plan to accept caraway if caraway, expected value is $.25

I suspect that the opposite is more common, that we suspect that we are not in love, or no longer in love, solely for the fact that evidence to the contrary, that we are in fact in love, is not a part of our evidence set.

Our errors in judgment are often grounded in these more fundamental errors. Another similar example is that we are not vulnerable to love or that we are invulnerable to love. What this means is that a person believes about themselves, that they are not of the sort who "is mushy" or "terribly affectionate", or that they are an extreme introvert, better off alone, etc. Perhaps this person may believe that they can thrive on short-lived, shallow relationships, or just as commonly, no relationship at all.

Analogously then, consider the following:

> Case: Stacked and Stuntin'. Suppose that you reliably outperform your level of confidence when it comes to identifying your desire states. For example, you regularly underestimate your ability to discriminate between desiring beef or chicken. Shown a sample of either, you will be wholly uncertain which you desire—you will assign equal credence to each. However, forced to choose, you do systematically better than chance at selecting beef when you want beef and selecting chicken when you want chicken—in each case, you select for the desire you have around 75% of the time. Moreover, you are certain that you are like this.

	Prior		Posteriors		
	[Marriage]	[Career]	Marriage	Career	
Marriage	$\frac{3}{8}$	$\frac{1}{8}$	P_{Marriage}	$\frac{1}{2}$	$\frac{1}{2}$
Career	$\frac{1}{8}$	$\frac{3}{8}$	P_{Career}	$\frac{1}{2}$	$\frac{1}{2}$

$$\begin{aligned}
&\text{Decline your choice:} && 0\\
&\text{Choose Marriage:} && \begin{cases} \$0.67 & \text{if Marriage;} \\ -\$1 & \text{if Career} \end{cases}\\
&\text{Choose Career:} && \begin{cases} -\$1 & \text{if Marriage; .} \\ \$0.67 & \text{if Career} \end{cases}
\end{aligned}$$

Maximize Act Expected Value: Decline bet in both states since expected value is -$.17

Maximize Plan Expected Value: plan to accept cumin if cumin and plan to accept caraway if caraway, expected value is $.25

This has implications for the kinds of decisions that we make. Examples like the one above illustrate a kind of decision problem in which an agent will not expect to do best by trying to follow a plan to maximize expected value. We make a decision thinking that we desire a given outcome. This can include for instance, that we may experience more joy later on by breaking up with our fiancé now because they demand too much of our time. Our time here being a direct and measurable result that can be quantified, for instance to answer the question "how much did I get done today?" On the other hand however, if we spent the day with a loved one, it might be less obvious how to answer such a question according to quantifiable metrics.

Self 1: "How much did I get done today?"

Self 2: "Not much, I just spent the day with my loved one".

Given our relationship with the other, "Not much" and "I just spent the day with my loved one" seem like contradictory statements. The more likely cause of this contradiction is that it is difficult to quantify one's time spent in this manner, but that is not the same as it is to say that it is difficult to qualify one's time spent in this manner. It may just be that rational choice theory does not have a way to qualify one's time spent even though it has a way to quantify one's time spent.

Often it turns out that the outcome that we thought we desired, was not the outcome that we desired should we achieve it. So while it would have been perhaps possible that the agent could have looked for evidence *that* $\neg p$, ***intentionally-biased inquiry, for instance, th****at maybe I am trauma bonding*, intentionally biased inquiry may be impossible The Externalist's Guide to Fishing for Compliments. However, intentionally-biased inquiry is not impossible. I argue here that cognitive technologies such as religious narratives represents a plausible intentionally-biased inquiry.

Eleonore Stump has argued that love is the interaction of two desires, the first for the good of the beloved and the second, union with the beloved. However, a person often suffers in that they are divided against themselves and so cannot experience desire for the good of the other, or desire for union with another. Desiring the good of, and union with, another places us into a vulnerable state and it is not always clear whether vulnerability in this way is a virtue. There is a lot of disagreement over what kinds of things we are, whether relationally vulnerable wherein we desire to be with another, or relationally invulnerable where the outcome of not being with another is not a value that we possess, or that it is not a value that overrides other values that we possess. Additionally, *internal integration of the will* would be necessary for love and union with another only if we are relationally vulnerable (Thomas Aquinas, "Adoro devote," in *The Oxford Book of Medieval Latin Verse,* ed. F.J.E. Raby (Oxford: Clarendon Press, 1959), 403-4) In part one, I focus on cognitive technologies such as historical narratives and consider their role in helping us to understand something about ourselves, whether or not we are the kind of persons who possess relational vulnerabilities. In part two, I attempt to use decision theory to understand how the question of vulnerability can impact our decision making abilities. I do this by focusing on how cognitive technologies, which are cultural means of emotion regulation

(De Cruz, 2024), impact our ability to make decisions when we are uncertain about our values, i.e., the nature of what makes us live well.

Eleonore Stump defends a two-part Thomistic account of personal identity. A person's first-order desires reflect the *reasons* she has to take something as good whereas their second-order desires reflect the kind of will they want to have. She then contrasts a metaphysics of personal identity which gives a special focus to the rational faculties that a person possesses, where a person's attribution of value is correlated with some particular action in some particular circumstance and a psychology of personal identity where a person's attribution of value is correlated with general reasons of action grounded in their value of a good will.

Awe is an epistemic emotion that we experience when we perceive or conceptualize vastness and need to cognitively accommodate our perceptions. One evening, yourself and some friends see an unfamiliar dirt road leading into what you know to be, for you, an unexplored area of the Southern California high desert. It's late, around 9pm, and this area is dark. You press on. The headlights from the car that you are driving give you an unwarranted sense of security amidst the pitch darkness of the desert around you. The road narrows somewhat and you notice a stark rise in elevation. There are several switchbacks as you continue to gain altitude. Finally the road ends at a small plateau which you ascend and find yourself on level ground. You get out of the car and turn off the car's lights looking up towards the clear night sky, at which point you have an unobstructed and unpolluted view of the night sky above. There's nothing to discover, or understand about what lies before you.

Similar to awe, wonder is also an epistemic emotion that we experience when we glimpse at something that is similar but still unknown to us. You are walking along a walking trail near your home when you see for the first time, an uncommon creature such as a snake of unusual hue and coloration. You stand in the same spot for several minutes while you try to think of all you have seen that is closely related to the object in front of you, and all that you might have heard.

Both emotions are important for the mental states related to knowledge. These include curiosity, doubt, surprise etc. It has been argued that either science or technology, can cultivate or threaten these experiences Wonderstruck: How Wonder and Awe Shape the Way We Think; Gabriel Marcel. There is a sense in which both of these emotions are important for special relationships. Parent, child, spouse, close friend, etc. It is also plausible to argue that these emotional states are synonymous with the state of being that we typically characterize as "in love". But before we provide a defense of this description of love, we should look at other possibilities.

Another way to understand this, is that the insights of science or meaningful lyrics as a cognitive technology, can help us to discover new sources of awe and wonder Wonderstruck: How Wonder and Awe Shape the Way We Think. De Cruz argues that cognitive technologies related to awe and wonder are a cultural means "to control our emotions" (Wonder, 14). In this way, cognitive technologies can either help or hinder our ability to make important decisions. For instance, scientists regularly use models to test their intuitions. Political scientists might build

a political model of a particular election in order to test their intuition about the viability of running one candidate over another. A military strategist might create a model of a military campaign to test their battle strategy.

Following Stump 2010, biblical narratives are cognitive technologies that enable deeper relationships with other people. This in turn allows us to make informed opaque decisions, decisions in situations where an agent is uncertain about their own credences. I attempt to test this claim by looking at an example of music lyrics by treating them as a kind of decision technology.

Some have argued that when we lose our sense of wonder, our sense of what defines us is lost. According to Gabriel Marcel, Gabriel Marcel; Gabriel (-Honoré) Marcel; Marcel, 1963; Marcel; as we look at another individual who would naturally represent a deep source of awe and wonder for us, we instead see a set of functions and sources of productivity. When our sense of wonder is lost, our sense of what defines us as humans is also lost.

Emotions are important for survival. They allow for adaptive responses to threats and opportunities in our environment, including social ones (Ekman 1992). Awe and wonder are *self-transcendent emotions,* they move our attention away from ourselves facilitating more adaptive group participation. This happens when these technologies can cause inappropriate emotional states. In other words, technology that may be the source of our failure to recognize the value in things that should be valuable. As humans, it is likely that we find special relationships essential for flourishing. Therefore, failing to recognize the value in special relationships is a failure to appreciate what is essential for human flourishing. Therefore science and technology undermines our sense of what defines us as human beings, such as we thrive in environments of mutual particular care.

Helen De Cruz, Helen De Cruz (St. Louis University), "Wonderstruck argues that this story about science is incomplete. The technologies in question either help us to make informed opaque decisions. One way they do this, is that the insights of science enable us to discover new sources of wonder. We are imperfectly reliable "at identifying our own credences, utilities, and available acts". According to Maximize Expected Utility, or Maximize Utility, we might then think that some songs as a technology can help or hurt our abilities to make informed decisions. Cognitive technologies serve human needs such as navigating our environment Wonderstruck: How Wonder and Awe Shape the Way We Think, 99. Therefore it is possible that cognitive technologies can serve human needs by helping us to support our credences. By treating lyrics instrumentally, commercial music production for instance, songs may help us to find or lose our sense of awe and wonder with regard to special relationships, they may help or hinder our ability to recognize the inherent value in this relationships by revealing to ourselves whether we are vulnerable creatures or not. For instance, a cognitive technology like the popular song "**By yourself",** might remove the pain behind a recent breakup. The importance of this is being in control of our thoughts and emotions, "controlling our emotions before they control us" On romantic love, 199.

The problem here though, is that while a technology like the song "**By Yourself"** can help us to control our emotions, they may do so by revealing to us that we are not in fact relationally

vulnerable. As such, we still need a way to determine whether we are relationally vulnerable or relationally invulnerable creatures. So songs may be the kinds of technologies that allow us to "control our emotions" when we are uncertain about our credences and expected values.

The first represents an emergent (like emergentist about mind) claim about the role of the science and technology in our loss of wonder, namely that the purpose of science and technology is to dispel our sense of awe and wonder (giving us control over our emotions) and what Helen De Cruz is saying, is that this view of science conflates the object, discovery, with a potentially indirect outcome.

Here, e.g., we might think that country music is a kind of technology that induces awe and wonder at special relationships, or mitigates awe and wonder on the prospect of a painful breakup, either of which may be adaptive. We can operationalize this outcome and ask whether there is a direct, or merely indirect relationship between science and apathy (a form of suffering). It warrants exploring to determine what each proposes of the relationship between scientific discovery, or awe and wonder technology and one's loss of meaning.

Other examples of cognitive technologies include video games. Both Star Citizen and Elite Dangerous are open world space simulators. They each use the "character regeneration" mechanic (respawning) to allow the player to continuing playing the game after a particular defeat without too much loss. For instance, a player who is transporting cargo from one planet to the next may be attacked by another player. Their character then dies. However, by introducing the re-spawn mechanic, the player's character regenerates somewhere else. So while they may have lost their cargo or their in-game loot, they can still continue to play.

Star Citizen's Chris Roberts has proposed a concept he calls "Death of a Spaceman". Accordingly, after too many death's of a player's character, the player's character would be reset, loosing not only any in game advances made up to that point, but also the character itself.

> I also feel that if everyone can be cloned easily, it fundamentally changes the structure of the universe. You now have a universe of immortal gods that can't be killed. Death is just a financial and time inconvenience that has no further consequences. The life and death cycle of humanity is what has brought us our history, our need to "make a mark" in our time, to push forward. If I want a living, breathing universe that has a lot of the dynamics of a real world and is inspired by the decline and fall of the Roman Empire, immortality for all is problematic. Death of a Spaceman - Roberts Space Industries | Follow the development of Star Citizen and Squadron 42

This proposal has been met with much criticism by gamers but deserves more engagement by philosophers. However the important point here is whether the re-spawn mechanic obscures our credences, what we are really like. Roberts here seems to think so. Once again, this is a question about whether we are vulnerable or invulnerable. Another proposal has been given by James Brown (Brown, James L. D. "How to Be a Prudential Expressivist." Mind 134, no. 534 (2025): 324–46. https://doi.org/10.1093/mind/fzae072.).

What we need to understand, is whether it is possible for cognitive technologies to help us determine whether we are relationally vulnerable or not. Some have argued that science and technology can cause us to lose our sense of awe and wonder leading to states of apathy. In such a state, our sense of what defines us" is lost. According to Gabriel Marcel, Rene Descartes, and Daniel Dennett, science and technology is responsible for this apathetic state. For instance, Daniel Dennett and Rene Descartes argue that the function of science is to remove the mysterious and that the mysterious is the source of awe and wonder. In the face of mystery, science is our attempt to uncover this mystery and thereby remove our sense of awe and wonder. When this happens, Gabriel Marcel; Gabriel (-Honoré) Marcel; Marcel, 1963; Marcel argues, we begin to only see ourselves as a set of functions and sources of productivity. This is because they make that which is mysterious, obvious. Helen De Cruz, Helen De Cruz (St. Louis University), "Wonderstruck argues that this story about science is incomplete. Rather, the insights of science enable us to discover new sources of wonder. Following De Cruz's description of cognitive technologies in her book *Wonderstruck*, either science or technology can mitigate threats to our personal sense of humanity. The insights of science actually helps us to discover new sources of wonder.

As an example, she uses what she calls religion as an "awe and wonder" technology. But do cognitive technologies either help or hinder our ability to make important decisions. Consider how cultural beliefs like the value of democracy persist in a culture. A person may have an idea, such as "democracy is an important social prescription and should be protected at all costs". The question then becomes how does this idea transmit throughout the culture? Religion too can support the transmission of important beliefs by making the idea uniquely memorable and therefore easy to transmit. To say that something is memorable in this way means , means that it becomes a new source of awe and wonder. A view of religion in the cognitive science of religion, is that cognitive technologies serve human needs such as navigating our environment Wonderstruck: How Wonder and Awe Shape the Way We Think, 99. There are two questions here; what are cognitive technologies, and how do they shape our ability to maximize expected value?

An example includes music lyrics. Decisions are not always about who we are. Country music for instance, often has been called authentic when portraying themes associated with family, home, traditional values (Richard Shusterman, "Moving Truth: Affect and Authenticity in Country Musicals," The Journal of Aesthetics and Art Criticism 57, no. 2 (1999): 221–33, https://doi.org/10.2307/432314.; John Dyck, "The Aesthetics of Country Music," Philosophy Compass 16, no. 5 (2021): e12729, https://doi.org/10.1111/phc3.12729.; Evan Malone, "Country Music and the Problem of Authenticity," Brit J Aesthetics 63, no. 1 (2023): 75–90, https://doi.org/10.1093/aesthj/ayac020.). Is this because a narrator is committed to their family, or because they *want* to be committed to their family? On the other hand, these songs may help us to find our sense of awe and wonder with regard to important pro-social values. For instance, when George Bush argued that "to listen to a country and western song is to hear the story of America set to music. It is a story of patriotism and hard work, a story of faith, opportunity, and achievement". It may be the case also however that commercial music production can also suggest value or the lack of it, in relationships. The technologies in question can either help us

to make informed decisions, decisions about future states of value where we would normally not have access. However, music is often controversial and our understanding of musical artifacts depends on our interpretation of the artist's values and credences. Open world video games on the other hand depend largely on the player's values and credences. In other words, unlike music, video games can enable to experience a value set assuming they immerse themselves in some given state s'.

Decision theory specifies which acts are rational given an agent's prior credences and utilities. However, sometimes an agent's prior credences are uncertain.

> "An opaque decision problem represents the situation of an agent who must decide what to do while being (potentially) uncertain about their own credences".

What that agent needs then is an opaque decision theory for when an agent has imperfect self-knowledge (uncertain about credences, utilities, and available acts). Normative theory of rational choice says that in deciding what to do we could either maximize expected value or maximize actual value. Maximize actual Value is often hard advice to follow. We are often uncertain about which acts available to us, will have the best outcome. How an agent acts depends on what plan they are trying to follow.

However, sometimes, they instead perform an act which their plan recommends at another state. Decision theorists often cite Bayesian statistics to model decision theories when their is a degree of uncertainty. However, we do not always know whether our doxastic states are rational or not, do not always have sufficient evidence. Therefore, orthodox Bayesians have a difficulty modelling a certain kind of uncertainty, when our credences are unknown (akrasia).

The following examples below represent two possible ways in which one should feel, but consider the guidance principal: "the probability of such an agent acting as if they are in a given state will be directly proportional to the credence they assign to being in that state" (Planning for Mistakes, 8).

The Predicaments

Below are two different predicaments, **Stunted Soul** and **Stackin' Cars**.

- "All this stuntin' couldn't satisfy my soul (–oul), Got a hundred big places, but I'm still alone (–one)"
- "Stacked your bread and bought your own Mercedes (Vroom, vroom). You your own boss, do it your way (Way)

In the first, the agent made a decision that they regretted. In the second, the agent made a similar decision as the first but did not regret it. In each, according to decision theory, "we are imperfectly reliable at identifying our own credences, utilities, and available acts". Does this fact explain the mistake made in the first predicament? Does the agent in the second predicament avoid this mistake? What is unique across each predicament is the probability

that the agent is in error regarding expected subjective value/desirability of the outcomes of their actions. For instance, the degree to which the agent in the first predicament is uncertain about her own credences, such as understanding how much value is owed and where.

Suppose that you are an agent who understands what some advice plan recommends in any possible circumstance and that you fully intend to follow the advice plan that will maximize you expected value. However, you do not know what your circumstances are and therefore you do not know what your "maximized expected value" will look like.

For opaque decision problem $\hat{\Delta}$: $\Pi = S \rightarrow A$ is the set of plans determined by $\hat{\Delta}$ where a plan is a function from states s, s',s'',s''',. . . $\in S$, and each s $\in S$, $\langle S, O, A, P - s, \upsilon \rangle$ is a standard Savage style decision problem that represents the agent's perspective at s at that world. Where a plan, $\pi, \pi', \pi'', ... \in \Pi$ are functions from states to acts. Given $\pi(s) = a$, we say that a is the act recommended by π in s.

Maximize expected value, $\Pi_{|MEV}$ is the set of plans to maximize expected value in $\hat{\Delta}$. $ev_s(a)$ is the expected value of a at s for any $s \in S$ and $s \in A$.

The expected value of an act form the perspective of s is the sum of the values of the outcomes produced by the act at each state where each is weighted according to the probability assigned to that state from the perspective of s.

$$ev_s(a) = \sum_{s' \in S} \upsilon(a(s')) \bullet P_s(s') \tag{1}$$

Given $\Pi_{|MEV}$, then $\pi \in \Pi_{|MEV}$ iff for all $s \in S$ and $a \in A$:

$$ev_s(\pi(s)) \geq ev_s(a) \tag{2}$$

Isaacson and Levinstein argue that decision theory can guarantee action in accordance with contingent "truths of which an agent is uncertain". We say that the agent is uncertain because in the first predicament, the agent admits that they were wrong when they believed they knew what was needed for a good life. For instance, that "a hundred big places" was sufficient for well-being. In the second, there are many people who may disagree that all a person needs for a good life is a Mercedes and the fleeting rendezvous, but the conclusion of this claim may be opaque. It is unclear whether the predicament is one in which the agent will still affirm her position in the future when other facts materialize or become more clear since our values are often more obvious when exposed to their consequences (Lasonen-Aarnio (2015), and has been discussed in Salow (2018) and Isaacs and Russell (2023)). It is not clear that once the agent has achieved the Mercedes and fleeting rendezvous, that they would still affirm their choice since the agent's values often only become clear when the agent is confronted with them.

Let's imagine that a person doesn't know how they feel about another individual. They are facing a moment wherein they must make a very important decision regarding their relationship with that other person. According to decision theorists, a decision theory can guarantee actions

that are contingent with truths that an agent is unsure of. What this means is that in cases where the agent is unsure of what would constitute a good life, decision theory may not have a way to calculate a rational and informed choice. What we need is to be able to determine the value of life with this other person or the value of life without this person.

Deciding on Love: The Logic of Decision

There are two options for deciding the rightness of an action in the face of epistemic uncertainty. According to consequentialism, the rightness or wrongness of an action is determined by that action's consequences without reference to the agent of the act. Therefore we should attempt to **Maximize Value.** We should do whatever it is that will result in the most valuable outcome. However, the weights that we apply to the outcomes of our actions are often dependent on the agent, whoever the outcome will have value for, the agent. But the agent herself is often unsure just how she will rate the perceived value of some given outcome before deciding how to act. Decision-Theoretic Consequentialism and the Nearest and Dearest Objection Therefore, since we are often unsure of our own preferences in this way, our epistemic access to the way the world is, is imperfect. As such, we should do whatever it is that will result in the **most *expected* value.** However, since expected value depends largely on what kind of thing the agent is, and this can at times largely be unknown to us, there should be some other course of decision metric available to us.

(Q2) a. What would agents need to be like in order to be able to expect to do best by trying to Maximize Expected Value?

b. Are we agents like that?

In other words:

Maximize Expected Value. Our credences are too opaque for us to always be able to identify the acts which maximize expected value (cf. Smith 2010; Srinivasan 2015; Lasonen-Aarnio 2019;2024; Lasonen 2025; Hughes 2021).

Example 1: Am I in Love?

What this question asks, is whether the agent is the sort of thing whose flourishing would be achieved by a given course of action, pursue a relationship, or to pursue wealth and status. In other words, the agent must decide on *life with* another person, or *life without* another person. Below we express the agent's choices using a probability matrix on the left and a desirability matrix on the right. Notice that we've assigned a probability of .5 to the agent being in love and spending their life with the person they are in love with, and .5 to not being in love and not spending their life with the person whom they are not in love with. We also assume what is desirable to the agent. We can assign numerical values to a given outcome, that represent the value an outcome may present to an agent. Say between 1 and -1 with 1 being the most

desirable and -1 being the least. E.g., being in love and spending your life with the person who you are in love with being the most desirable state and not being in love but spending your life with the person whom you are not in love with being the least desirable state. We then run a formula which gives us a product of the probability for some given outcome, against the value we might assign that outcome and then sum across the row.

Probability

Desirability

	In Love	Not in Love
Life with	.5	.5
Life Without	.5	.5

	In Love	Not in Love
Life With	1	-1
Life Without	0	1

Desirability of Life With: .5 + -.5 = 0

Desirability of Life Without: 0 + .5 = .5

What we end up with is an average value of life with a significant other at 0 while the average value of life without a significant other is .5. As such, we should follow a life plan that does not include life with a significant other since the risk that we end up in out of love/out of love is too high.

Given that we do not know whether or not we will value a future life with this person, it seems that the desirability of a life with them is net neutral whereas a life without them is net positive, .5. Here it seems that the best position is to pursue a life without any other person since living life without the person you are in love with affords a higher chance of success. The problem with this position is that it appears counter intuitive of the predicaments expressed earlier. Though it will be important to primarily focus on the first predicament, it will also be helpful to mention here that we can take the second to be a prior state of the first, with the second being a realization that the credences in the first were held in error. For instance, that the agent's degree of belief that a Mercedes and periodic rendezvous were sufficient for a good life. If the agent is not in love, then there is a higher likelihood that a Mercedes and periodic rendezvous are sufficient for a good life. If an agent is in love, then these would not be sufficient for a good life.

Another question is whether an agent is such that being in love, or having valuable, vulnerable relationships is a necessary feature of a good life. Eleonore Stump, Martha Nussbaum, Alasdair MacIntyre have referenced arguments to this effect (Martha C. Nussbaum, The Fragility of

Goodness: Luck and Ethics in Greek Tragedy and Philosophy, 2nd ed. (Cambridge University Press, 2001), https://doi.org/10.1017/CBO9780511817915.; Eleonore Stump, Wandering in Darkness: Narrative and the Problem of Suffering (Clarendon Press ; Oxford University Press, 2010); Alasdair C. MacIntyre, Dependent Rational Animals: Why Human Beings Need the Virtues, The Paul Carus Lectures (Open Court, 1999)). If it is, then a Mercedes and periodic rendezvous are not sufficient for a good life whereas if the agent is such that relationships which make them vulnerable is not a necessary feature of a good life, then a Mercedes and periodic rendezvous are sufficient for a good life. Each predicament also holds a given value for an agent. This value represents how satisfied the agent is with that predicament Causal decision theory.

- "Stacked your bread and bought your own Mercedes (Vroom, vroom). You your own boss, do it your way (Way)
- "All this stuntin' couldn't satisfy my soul (–oul), Got a hundred big places, but I'm still alone (–one)"

Rearranging the predicaments allows us to notice the opacity of the agent's values more clearly. Here the agent only realizes too late that she does place more value in living a life with the person whom she is in love with, than she does in living a life with the person whom she is not in love with (whether alone or in shallow and fleeting relationships). We might then suppose that the agent is vulnerable in a special way. To understand this better, we imagine two possible value states, vulnerable and invulnerable. In ***The Fragility of Goodness***, Martha Nussbaum argues that a special trait of humanity is vulnerability in that 1) flourishing requires that human beings have special relationships with others, 2) that in these relationships they must be vulnerable such that were these relationships to fail in some way, the person would be destroyed.

Example: Vulnerability

	Vul	Inv
Life with	.5	.5
Life Without	.5	.5

	Vul	Inv
Life with	1	0
Life Without	-1	1

(.5)(1)	(.5)(0)
(.5)(-1)	(.5)(1)

.5	0
-.5	.5

Vulnerable:

Desirability of Life With: .5 + -.5 = .5

0 Desirability of Life Without: + .5 = -.5

But the difference between being vulnerable and being invulnerable is stark:

Invulnerable

Desirability of Life With: .5 + -.5 = 0

Desirability of Life Without: 0 + .5 = .5

What does this tell us? That assuming we are vulnerable creatures, we are best to pursue special relationships and that if we are invulnerable creatures, then we are best to avoid pursuing special relationships. What remains to be seen then, is given that we might recognize that the question of whether or not we are vulnerable creatures is in itself opaque. Are we, or are we not vulnerable creatures? Is any one person vulnerable while others are not? How do we determine whether or not we are vulnerable? The agent in ***Stacked*** thought they were invulnerable, but realized only too late that they were in fact vulnerable. We need a way to decide between ***life with*** and ***life without*** when we are uncertain about whether or not we are in fact vulnerable or invulnerable. If we are in love, then we might think that we are vulnerable since being in this state implies that we are the kinds of creatures for whom being in love is a great good. If we are not the sort of creatures such that being in love is a great good, then we might wonder whether or not we are invulnerable and whether not being in love implies that we are not the kinds of creatures for whom being in love is a great good.

Does opaque decision theory help us to solve this problem? Sam Carter recognizes a problem in maximizing expected value. The question turns back to whether or not we are a certain type of agent.

- (Q1) Can an agent always expect to do best by trying to follow Maximize Expected Value?

I will argue that, when taken in full generality, the answer to this question is:

No. Agents who are prone to making mistakes of certain kinds should sometimes expect to do better by trying to follow other advice instead. This leads to two further questions:

- (Q2)

    a. What would agents need to be like in order to be able to expect to do best by trying to Maximize Expected Value?

    b. Are we agents like that?

It seems that according to Carter, agents would need to be of the sort that they are integrated in will.

For Instance:

These are states where the agent is assumed to be certain about which acts are available to them and what value they assign to the outcome of those acts. For instance, each line below assumes a credence, agent believes that life with material possessions is conducive of a good life and come to the realization that they are not agents like that.

> We are not always certain what the world is like. For this reason, Maximize Expected Value advises us to act according to what we take the world to be like, instead. However, trying to follow its recommendations can leave us predictably worse off. **Sometimes, we are not certain what we take the world to be like, either (am I in love?).** And when we are not certain what we take the world to be like, we may fail to do what it tells us to do. For agents like us, susceptible to human types of ignorance and error, the advice to maximize expected value will not be good, fully general advice for decision-making.

Reworking the Argument

Basic Framework for Agents who are uncertain about their credences

Definition 1

Opaque Decision Problem

$\hat{\Delta} = [S^{\hat{\Delta}}, O^{\hat{\Delta}}, A^{\hat{\Delta}}, P^{\hat{\Delta}}, v^{\hat{\Delta}}]$

As such, we have a decision to make. This decision is represented by the formula above which says the following:

A set of worlds, Ω, a world consists of various possibilities. All this means to say is that there are some occurrences which are possible at one world that may be impossible in another. For instance, a world in which Superman's father passed away before meeting his mother is a world where Superman's existence would be impossible. A set of States, S, where an agent is perhaps in love or out of love. A set of outcomes, O (life of flourishing, mental well-being, emotional-being, relational-wellbeing, fulfillment, well-adjusted children who are emotionally healthy, . . .). An outcome can include an agent who goes onto become a professional athlete

or a long-haul truck driver. A set of acts, $A \subseteq S \rightarrow O$ where if an agent performs an act while in a given state, then they can expect a particular outcome. For instance, an agent might work more and spend less time with family/significant others

- A state-dependent probability function, $^{**}{*}P : S \rightarrow (P(S) \rightarrow [\ 0,\ 1\ ])$, (**What is the probability that the relationship that you sacrificed the college football scholarship will work out?)*
- A value function, $v : O \rightarrow \mathbb{R}$**,** (how do we determine the value of a state given a set of outcomes?)

The objects of the agent's uncertainty

$^{**}{}_{S}states, s, s', s'', s''' \ldots \in S_^{**}$ include objects of agent's uncertainty, i.e., I may be in love or relationally vulnerable

> States settle everything about the way things are which is relevant to the agent's decision and about which they may be uncertain. This will include facts about the external world (e.g., the number of marbles in the jar) as well as facts about the agent's internal life (e.g., what credence they assign to hypotheses about the number of marbles in the jar).

The agent may be uncertain whether or not they are truly in love. If they are truly in love, then the value of being in love will outweigh any potential (super)material (beyond what an agent needs to survive). This is the state of affairs in predicament ***Stunted Soul***.

An agent desires a particular outcome

$Outcomes, o, o', o'', o''', ... \in O$

> Outcomes settle everything about the way things could be which is relevant to the agent's decision and which the agent may value differently (e.g., whether they win the marble guessing game).

Here, the agent recognizes the outcome of her action, that she was in fact vulnerable.

The Agent's acts

$Acts, a, b, c... \in A$

> Each act is a function from states to outcomes. They represent the basic acts which are—in some sense—under the agent's control (e.g., guesses about the number of marbles in the jar).

The agent chose to forgo the relationship in order to pursue an expensive car.

Credence Distribution

We have a set of state-dependent probability functions. Each set of state-dependent probability functions represent the agent's credences (believability, how believable (e.g., life with only material possessions) it is for the agent) at each state.

$s \in S, P_s$

What is the probability that the agent is relationally vulnerable?

υ = equals how well the agent takes things to be going. E.g., being covered in carats, or having mahogany cabinets.

The value the agent would place on the outcome.

An opaque decision problem represents an agent's perspective on the world, how they understand the world to be, and on their own credences, how they understand themselves to be, whereas a savage-style decision problem represents an agent's perspective on the world only. Below we can look at some charts that represent possible actions and outcomes.

Take the money and run

	Actions	
	More Money	Less Money
More Time with Kids	-1	1
Less Time with Kids	1	-1
Take Job in Foreign Country	1	-1
Stay close to family	-1	1
. . . e.g.,		

To be in love or not to be in love

	Actions	
	In Love	Not in Love
More Time with another	-1	1
Less Time with another	1	-1
Take Job in Foreign Country	1	-1
Stay close to other	-1	1
. . . e.g.,		

The tables below represent an agent's plans to act:

First recall the two predicaments. Notice that I have left probabilities and desirability blank for now. Each set represents the plans an agent can make to achieve well-being.

The agent chooses wealth over relationship: Stackin' Cars

Probability

	Money	No Money
Love	.5	.5
No Love	.5	.5

Desirability

	Money	No Money
Love	1	-1
No Love	1	0

Stunted Soul'

But then the agent regrets their choice because they didn't realize that they would have valued relationship. This is because the agent's future values were opaque.

Probability

	Money	No Money
Love	.5	.5
No Love	.5	.5

Desirability

	Money	No Money
Love	1	0
No Love	-1	1

Plans

Notice too that for any opaque decision problem Δ:

$\Pi = S \rightarrow A$ is the set of plans for agent in Δ. A plan then might be that if an agent is in state S, say $s, s', s'', ...s_n \in S, P_s$, say some state of vulnerability, then it is probable that agent may pursue any number of actions $a, a_1, a_2, ...a_n \in A$ etc. Some states that are important here include having money versus being broke, being in love and in a relationship, being in love and not being in a relationship, not being in love but being in a relationship, not being in love but being in a relationship and not having money etcetera. The question is which of these actions, and their subsequent states, will produce what that agent will recognize as valuable when at a future state?

Plans then equate to a particular function from some given state to some given action

$\pi, \pi', \pi'', ... \in \Pi$.

As such then,

$\pi(s) = a$: means that act a is recommended by plan π in state s.

What does this look like then when the agent attempts to ***maximize expected value***? It depends on what state the agent is in, whether they are in love or not. Notice too however, that there is an implication about whether it is possible for the agent to actually be in love. If the agent is vulnerable, then they can be in love. But if they are invulnerable, than it is unlikely that they are in love. As such, in order to maximize expected value, we need a formula that can enable us to determine whether the agent is of the kind that being in love is a necessary and sufficient condition for well-being.

Maximize Expected Value

MEV characterizes a set of plans Π that will maximize expected value in opaque decision problem Δ. $\Pi_{|MEV}$, $ev_s(a)$ is the expected value of a at s for any $s \in S$ and $s \in A$.

Definition 3

$$ev_s(a) = \sum_{s' \in S} v(a(s')) \cdot P_s(s')$$

We can understand this as a claim about how an agent may attribute value to some given action based on the state that she is in. For instance, in ***Stackin' Cars***, the agent is fairly satisfied with her life. However, if the agent is the same agent as the agent in ***Stunted -oul,*** then it appears that she at some point become loses her satisfaction. ***Stakin' Cars*** created a course of action that included a particular plan. This plan was created according to which some action ***a*** would have an expected value ev given state ***s.***

Therefore, the expected value of the action given a state is equal to the sum of one of those states in a set of states, and the value of the action performed when the agent is in that state. We multiply this by the probability that that state is one of the states in the set. The expected value of some *act a* at *state s* is the sum of the values of the outcomes produced by that act at each state. What may help us to understand this, is to imagine the character in ***Stackin' Cars*** trying to decide whether she should pursue the record label contract, or pursue the significant other. She needs to determine what the expected value of each action would be for her in a future state, all while not knowing what this future state will be.

If this is the case, then some plan will maximize expected value though that plan which is in a set of plans iff for all $s \in S$ and $a \in A$:

$ev_s(\pi(s)) \geq ev_s(a)$

Therefore, the expected value of some plan that considers state of the agent s, is greater than or equal to the expected value of action a given state s. The question has to do with how do we determine the expected value of an action for the agent when the agent is in state ***s***. What we are expecting to determine is the sum of the values of each of the outcomes produced by an action when the acter is in a given state. We weight this according to an assigned probability that the actor will be in that that state, which tells us the value of the action for that agent when they are in that state. A good plan then would maximize the value expected in each state for one of the actions that have optimal expected value at that state. In summary, the expected value of some state recommended by some plan for that state is greater than the expected value of an action of some state.

Maximizing Expected Value

Remember that at the beginning, we questioned whether an agent ought to maximize value, do whatever has the most valuable outcome, or whether they ought to maximize expected value, do whatever has the most expected value.

- As a result, upon trying to follow its recommendation, we will often fail in the same way the agent in SC did. Therefore we should try to maximize our expected value, that is, to maximize the values we expect to have at some future state.
- But too: For agents like us, susceptible to human types of ignorance and error, the advice to maximize expected value will not be good, fully general advice for decision making. The agent in SC failed in trying to maximize expected value.